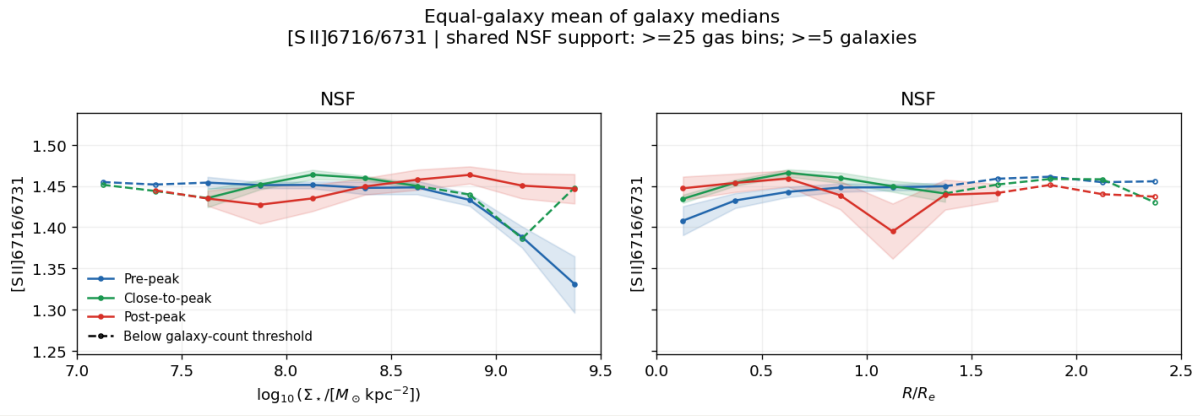
20261018 SII Ratio and EWHa in NSF across Infall Stages

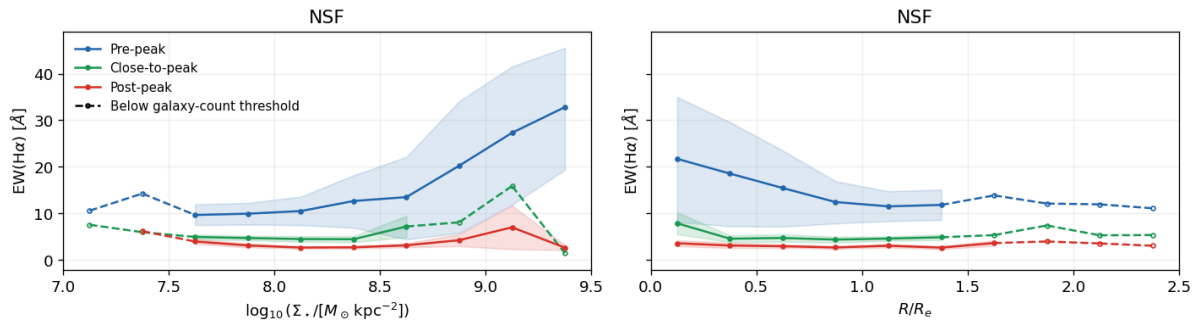
Dashed lines are the regimes where each bin has less than 5 galaxies, so I label them as "incomplete" data by still show their trends.

So we can see all NSF regions are almost reaching the saturation limit (low electron density) of [S II] doublet ratio, but for $\log\left(\Sigma_*/M_\odot \mathrm{kpc}^{-2}\right) > 8.5$, the pre-peak sample shows the decreasing trend (higher electron density) toward 1.33. Although 1.33 corresponds to $n_e \sim 100 \mathrm{cm}^{-3}$ under the assumption of $T_e = 10000K$, which is still roughly near the insensitive range of [S II] doublet ratio (Kewley+2019), I think this suggests the following signal: the high-mass end pre-peak NSF are potentially associated with a stronger contribution from leaked O/B photoionisation close to SF complexes, as what Baron+2026 found in PHANGS-MUSE, and so this support the relatively denser ionised structures/envelopes and less BPT line ratios here; while after RPS in post-peak, the dimishing of leaky photons revealing an increasing relative contribution from low-density, diffuse emission, potentially powered by HOLMES, and so this high-mass regime becomes lower electron density, lower EW, harder spectrum and potentially higher dispersion.

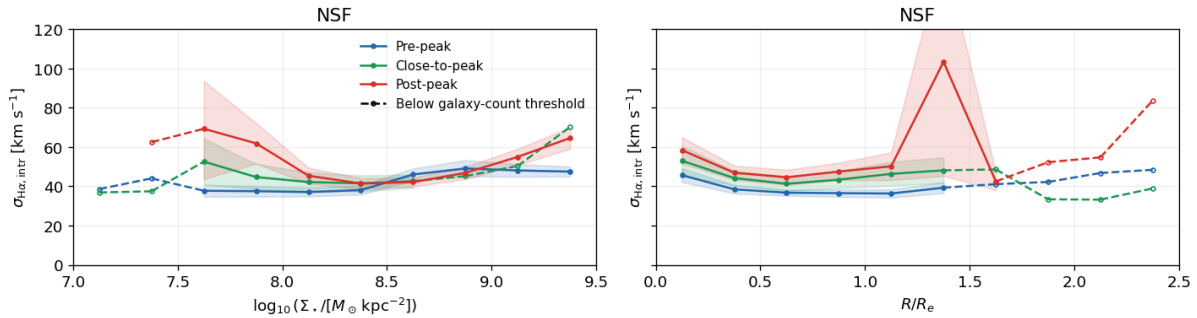
As usual, we show the equal-galaxy weighted results and pool-spaxel results.



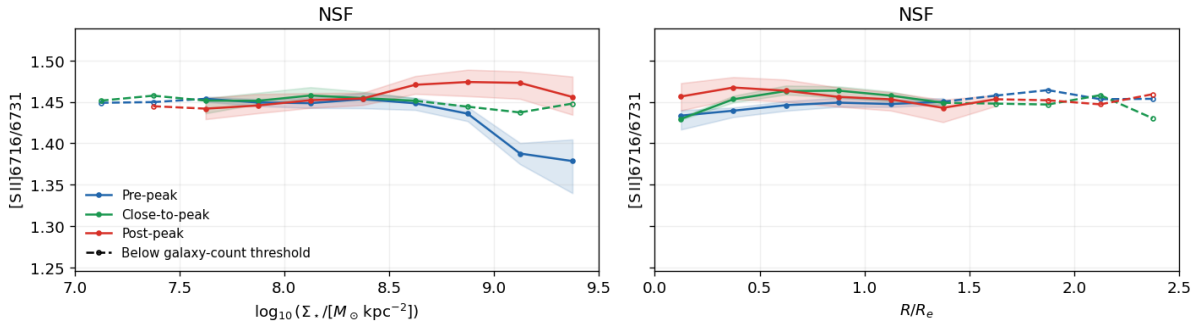
Equal-galaxy mean of galaxy medians
EW(Hα) [Å] | shared NSF support: >=25 gas bins; >=5 galaxies



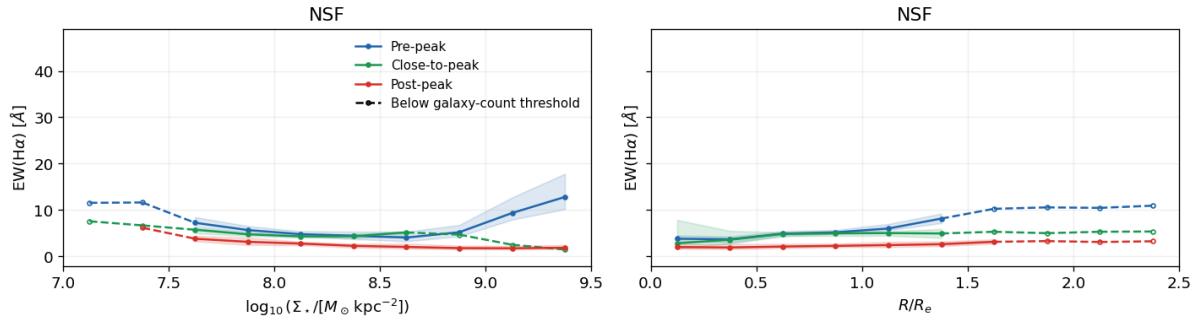
Equal-galaxy mean of galaxy medians
$\sigma_{H\alpha,\,intr}$ [km s^{-1}] | shared NSF support: >=25 gas bins; >=5 galaxies

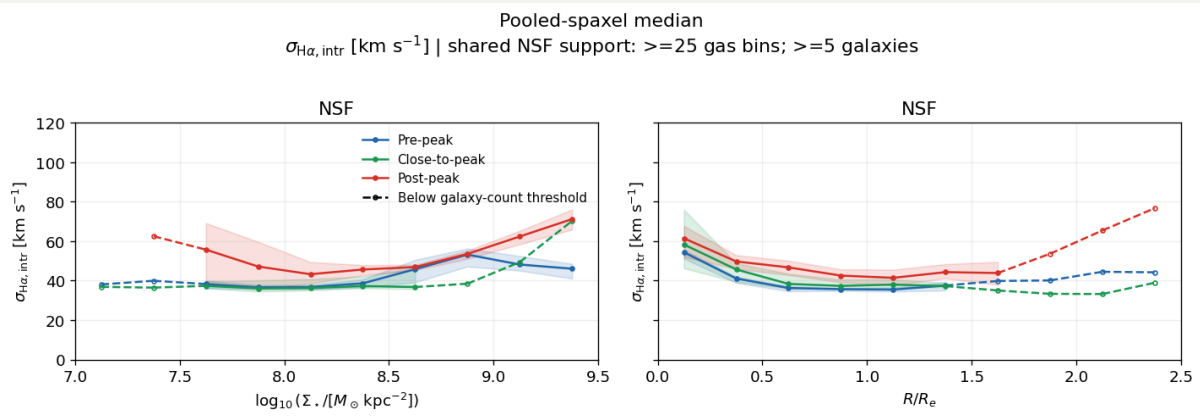


Pooled-spaxel median
[S II]6716/6731 | shared NSF support: >=25 gas bins; >=5 galaxies



Pooled-spaxel median
EW(Hα) [Å] | shared NSF support: >=25 gas bins; >=5 galaxies




Pooled-spaxel median
$\sigma_{H\alpha,\,intr}$ [km s^{-1}] | shared NSF support: >=25 gas bins; >=5 galaxies
NSF
Pre-peak
Close-to-peak
Post-peak
Below galaxy-count threshold
$\sigma_{H\alpha,\,intr}$ [km s^{-1}]
$\log_{10}(\Sigma_\star/[M_\odot\,\mathrm{kpc}^{-2}])$
NSF
$\sigma_{H\alpha,\,intr}$ [km s^{-1}]
R/R_e